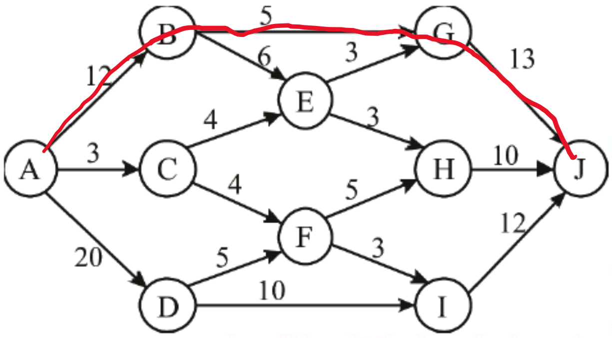
1. Show the execution of the Edmonds-Karp algorithm on the given flow network (source: A and sink: J).



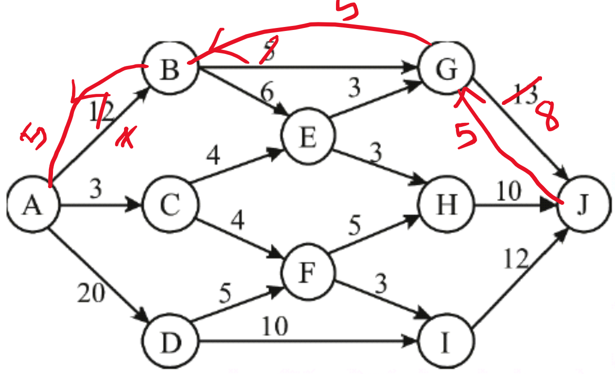
The **Edmonds-Karp algorithm** is an implementation of the Ford-Fulkerson method that uses **Breadth-First Search (BFS)** to find the shortest augmenting path (in terms of number of edges) from the source (A) to the sink (J).

Execution Steps

We start with zero flow and find augmenting paths until no more paths exist in the residual graph.

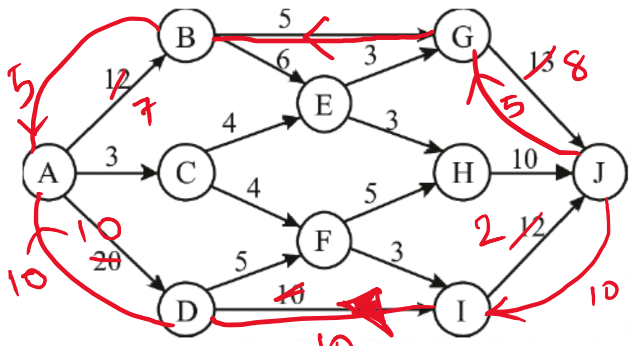
Iteration 1: Path A → B → G → J

- **Path edges/capacities:** (A,B):12 , (B,G):5 , (G,J):13 .
- **Bottleneck capacity:** minimum(12, 5, 13) = {5} .
- **Update:** Residual capacities decrease by 5. **Total Flow = 5.**



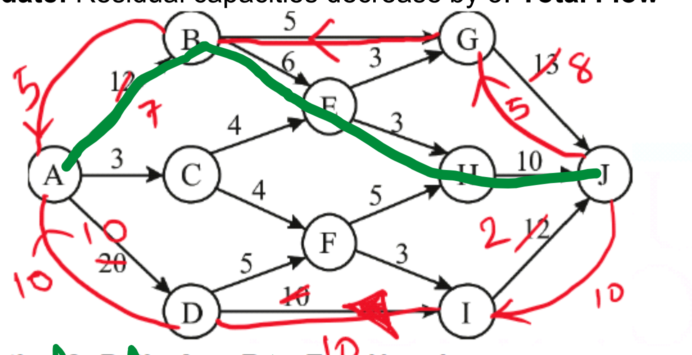
Iteration 2: Path A → D → I → J

- **Path edges/capacities:** (A,D):20 , (D,I):10 , (I,J):12 .
- **Bottleneck capacity:** minimum(20, 10, 12) = {10} .
- **Update:** Residual capacities decrease by 10. **Total Flow = 15.**

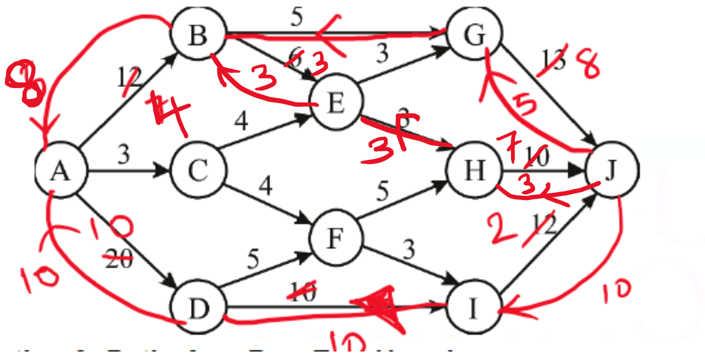


Iteration 3: Path A → B → E → H → J

- **Path edges/capacities:** (A,B):7 (rem.), (B,E):6 , (E,H):3 , (H,J):10 .
- **Bottleneck capacity:** minimum(7, 6, 3, 10) = {3} .

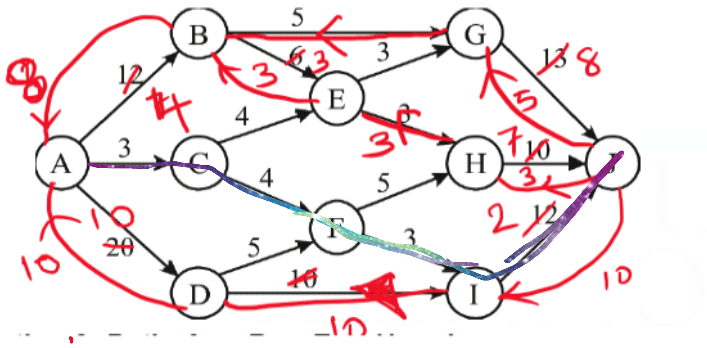
- **Update:** Residual capacities decrease by 3. **Total Flow = 18**.
- 

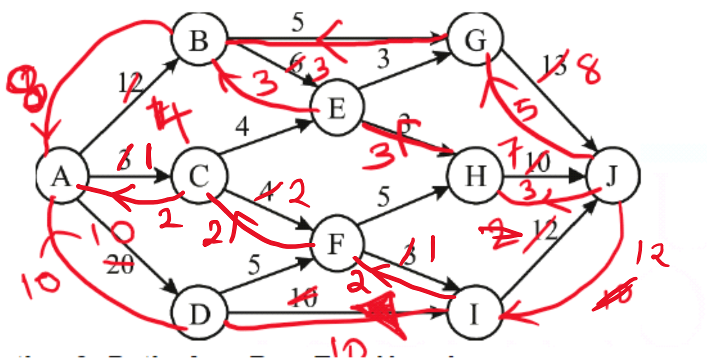
Residual graph ↓



Iteration 4: Path A → C → F → I → J

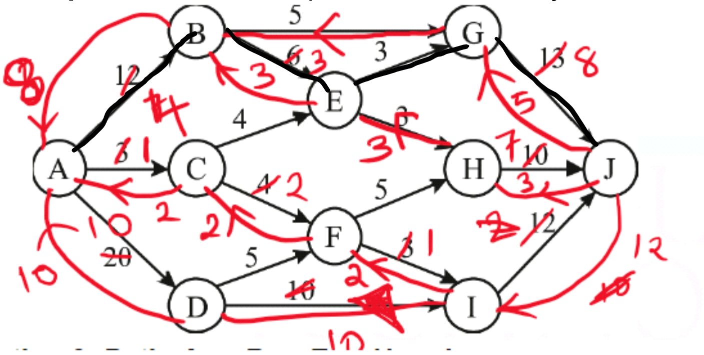
- **Path edges/capacities:** (A,C):3 , (C,F):4 , (F,I):3 , (I,J):2 (rem.).
- **Bottleneck capacity:** minimum(3, 4, 3, 2) = {2} .
- **Update:** Residual capacities decrease by 2. **Total Flow = 20**.

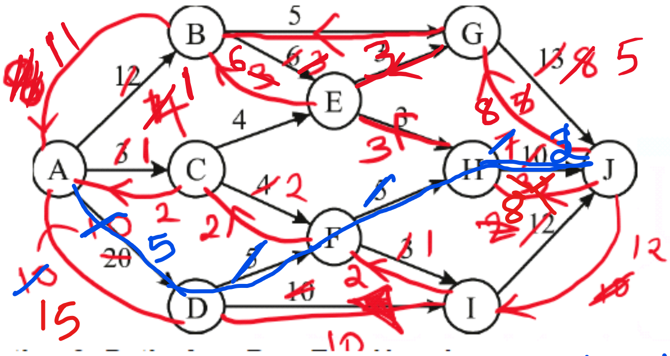




Iteration 5: Path A → B → E → G → J (Next shortest available)

- **Path edges/capacities:** (A,B):4 (rem.), (B,E):3 (rem.), (E,G):3 , (G,J):8 (rem.).
- **Bottleneck capacity:** minimum(4, 3, 3, 8) = {3} .
- **Update:** Residual capacities decrease by 3. **Total Flow = 23**.





flow =
23
Taking the path ADFHJ
flow = 23 + 5 = 28
which is the maximum flow.

Final Result

After these iterations, let's check remaining capacities from source A :

- A → B : 1 remaining.
- A → C : 1 remaining.
- A → D : 10 remaining.

However, the bottlenecks downstream (like E → H , E → G , I → J , and H → J) are now saturated or lack a valid BFS path to J .

Max Flow = 23